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Workbook Course 1

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Lecture 1: Optimising nutrition for performance

Section 1: Nutrition Professionals

Task: From the lecture, distinguish between the three roles below. Fill in the table.

Role	Qualification/Regulation	What they can do	What they cannot do	Example in Sport Context
Dietitian	Qualified healthcare professionals.	Interpret the science to support patients.	Cannot offer advice where there is personal financial gain.	
Nutritionist	do provide information about food & healthy eating.	educate, educate, write and healthy issue.	cannot provide info about special diets.	sports nutrition on SENR registration
Nutritional Therapist	use of vitamins for diet & lifestyle. to prevent ailments.	Allowed to sell supplements to their clients.	Cannot use alternative treatments do 'front' products without evidence.	design plan for someone who want to remove digestive tract health

providing advice for basketball player in fitness center.

educate

make food plan for professional soccer team.

Section 2: Why is Nutrition Important?

Questions: "Good nutrition won't make an athlete great, but poor nutrition can make an elite athlete average."

1. In your own words, why is tailoring nutrition important for athletes?

tailoring for athletes could help ~~see~~ the athletes fix the problem in certain way, to provide best performance

2. What might happen to an athlete's performance if their nutrition is poor?

poor repriment, hard to keep up training, low energy level, store too much adipose (fat).

Section 3: Macronutrients

Fill-in-the-blank exercise:

- The body stores approx. 100 g glycogen in the liver and 400 g in muscle cells.
- Protein is made of amino acid and is used for build up cells. ← enzymes, hormones, antibody.
- Fats provide essential fatty acid and fat-soluble vitamins like A, D, and E.
- Case study: An endurance runner avoids carbohydrates because they want to 'stay lean.' What risks might this pose for their training and performance?

Carbohydrate is essential for brain & central nervous system, avoiding carbohydrates might cause dizziness, undernourishment and low level of blood sugar.

Section 4: Energy Requirements

Step-by-step activity: Using the Harris-Benedict Equation, calculate your own RMR.

Formula (metric):

Men: $(10 \times \text{weight in kg}) + (6.25 \times \text{height in cm}) - (5 \times \text{age in years}) + 5$

Women: $(10 \times \text{weight in kg}) + (6.25 \times \text{height in cm}) - (5 \times \text{age in years}) - 161$

- Calculate your RMR: $10 \times 81 + 6.25 \times 185 - 5 \times 21 + 5 = 1866.25$

1966.25

1.5

3. Add exercise calories: $170 \times 1.5 = 255$ kcal/day

2969.375

Section 5: Hydration

1. List three signs of dehydration.

Headaches + dizziness. dry mouth. Poor concentration.

2. How does dehydration affect performance (heart rate, body temperature, cognition)?

Heart rate (111 = 81bpm) Body temp. (111 = 0.3°C)

3. Sweat Rate Activity:

- Record pre-session weight: 74 kg

- Record fluid consumed: 1 L

- Record post-session weight: 74 kg

- Fluid loss = 1 L

- Sweat rate per hour = 1.5 L/hr

Section 6: Types of Fluids

Fluid Type	Description	Example
Hypotonic	Low ion concentration ($< 2\%$)	Water
Isotonic	Same concentration as body fluids.	Muscle recovery (energy drink)
Hypertonic	High carbohydrate level ($> 10\%$)	Protein / Electrolyte

Section 7: Application to Sport Rehabilitation

Case Study: Basketball Ankle Injury

A 20-year-old basketball player has suffered a severe ankle sprain and will be unable to play or train fully for 6-8 weeks. They are worried about losing fitness and gaining unwanted body fat during recovery.

1. What are this athlete's key nutritional priorities during rehabilitation?

Protein, high quality food. Protein. Healthy fats & complex carbohydrates.

2. What foods or nutrients may specifically help with reducing inflammation and aiding tissue repair?

Protein & amino acids (bone broth / chicken). Vitamin C: collagen formation. Fruits & vegetables.

3. How would hydration still play a role in their recovery, even though they are less active?

Protein metabolism / keep the digestive system healthy (condition of).

Section 8: Reflection

- One new thing I learned today: general inhibition, / drinks class
- How I will apply this knowledge in practice: eat, play
- One area I want to research further: hydration / physical chemistry

hydration / physical change
Relative p.

Lecture 2: Macronutrients and micronutrients

Section 1: Macronutrients

1. Carbohydrates are described as 'TO GO'. What does this mean in terms of their role in performance?

Carb provide body with quick & primary energy

2. Why do glycogen stores need to be replenished after exercise?

During exercise muscle & liver glycogen is used as fuel.
After, it needs to be replenished.

3. Explain the difference between wholegrain carbohydrate sources and refined carbohydrate snacks for athletes.

slowly digested steady provide energy ; quickly digested.

4. Fill in the table below with examples of carbohydrate sources:

Type of Carbohydrate	Examples
Fruit	Bananas, apples.
Snacks	Granola bar, rice cakes, crackers.
Meal options	Pasta, rice, potatoes,
Sports products	Sports drinks, energy gels.

5. Protein is described as 'TO GROW'. Why is protein essential for athletes?

Protein → repair & growth. and immune function.

6. What are the 9 essential amino acids, and why must they come from the diet?

Histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, Tryptophan, Valine.

7. Fat is described as important for brain, muscle, and cell function. List three reasons why fat is important in an athlete's diet.

provide long lasting energy / supports hormone production & cell function / aids absorption. (A, D, E, K)

Section 2: Fuel for the Work Required

Questions:

1. How do carbohydrate requirements vary between easy training, moderate training, and game day?

less moderate high take.

2. What is carbohydrate rinsing, and in what scenarios might it be useful?

Carb rinsing

Section 3: Key Micronutrients

Questions:

1. Why is iron important for athletes, and which groups are at higher risk of deficiency?

essential for O_2 transport in the blood (haemoglobin), energy production, and reducing fatigue. 1) Vegans 2) Athletes with high sweat loss

2. List three food sources of haem iron and three of non-haem iron.

3) Endurance athlete

haem
non

Red meat Poultry Fish
Lentils Beans fortified cereals

3. What role does calcium play in bone health and recovery?

Calcium supports strong bones and teeth, muscle contraction, nerve function, and plays a key role in bone repair and adaptation from training. It helps prevent stress.

4. Why might athletes on plant-based diets need to consider vitamin B12, vitamin D, zinc, iodine, and omega-3 fatty acids?

plant-based diet will lack of certain nutrients mainly found in animal foods.

Section 4: Plant-Based Diets

Task:

You have an athlete following a vegan diet. What additional considerations do you need to think about in terms of protein and micronutrients?

Fill in the table with possible plant-based food sources:

Nutrient	Plant-based Sources
Protein	<u>Tofu</u> <u>lentils</u> <u>at yogurt</u>
Vitamin B12	<u>Fortified plant milks.</u>
Omega 3 fatty acids	
Calcium	<u>milk</u> <u>cheese</u> <u>tofu</u>
Iron	<u>Beans</u> <u>lentils</u> <u>spinach.</u>

Section 5: Recovery Nutrition

1. What are the 4 R's of recovery nutrition, and why are they important?

Refuel Repair Rehydrate Rest.

2. Give an example of a post-exercise snack that combines protein and carbohydrate.

Yogurt with banana.

3. What are some common signs of overtraining, and how might poor recovery contribute to these?

headaches, low mood, injury risk

Section 6: Sleep and Nutrition

1. Why is sleep quality and quantity particularly important for athletes?

physical recovery cognitive performance. hormonal balance.

2. Give two examples of foods that may improve sleep quality and explain why.

Kiwi fruit. → high in serotonin

milk → protein improve sleep quality.

Section 7: Nutrition for Injury

1. Why might energy needs increase during injury, even if training volume is reduced?

help tissue repair & inflammation control

immune response preventing muscle loss

2. What protein intake is recommended during injury recovery, and how should it be distributed across the day?

3. List key nutrients for bone injuries, tendon/ligament injuries, and muscle injuries.

Calcium Vit D

collagen vit C, protein

creatine.

protein, vit D, omega-3 fatty acid.

4. How might nutrition help with concussion recovery?

protein & micronutrients → help neural repair.

Anti-inflammatory food: Omega-3 fatty acid.

Section 8: Reflection

• One key learning point from today:

• How I will apply this in practice:

• One area I want to explore further:

Lecture 3: Dietary assessment and supplements

Section 1: Nutritional Assessment

1. What are the four main components of a nutritional assessment (A, B, C, D)?

Anthropometry Biochemistry Clinical Dietary & Everything else.

2. What are two limitations of using BMI for athletes?

Does not distinguish muscle from fat ; Does not reflect fat distribution

3. Why might measuring body composition be important for athletes?

To monitor training progress, to optimize performance, manage health risks.

4. List three biochemical markers relevant to sports nutrition.

Iron status — Ferritin Protein status — Serum albumin.
Inflammation — Creatine kinase.

Section 2: Taking a Diet History

1. What are the main types of diet history methods (e.g. 24-hour recall)?

24-hour dietary recall / Food frequency questionnaire / Food diary: record.

2. What are the pros and cons of food diaries compared to dietary recalls?

pros: Detailed & real-time Quick
cons: may change behaviour Memory-based, may miss details.

3. What information is important to collect about an athlete's daily intake (beyond food type)?

Portion sizes & cooking methods Meal timing & distribution around training.

4. Practice Task: Interview a peer and take a short diet history. Record:

- Breakfast:

- Lunch:

- Dinner:

- Snacks:

- Drinks:

Section 3: Supplements

1. How common is supplement use among athletes?

60% - 80% / 80%

2. What is the principle of 'strict liability' in relation to supplements and anti-doping?

Strict liability means athletes are responsible for everything they ingest.

3. What steps should athletes take before using supplements?

Assess need with a qualified sports nutrition & check product safety;
Evaluate evidence for safety & third party testing.

Section 4: Why Athletes Take Supplements

1. List three reasons why athletes might choose to take supplements.

To enhance performance; To support recovery and training adaptation;
To correct or prevent nutrient deficiencies.

2. What are two potential concerns with supplement use?

Reverse health risks (overdose) & Some supplement may have side effect.

Section 5: Types of Supplements

Task: Fill in the table below with examples and uses.

Type of Supplement	Examples	Purpose/Use
Sports Food	sports drinks, gels, protein powder	provide convenient fuel, hydration, or protein around training.
Medical Supplements	Iron, Vit D, Calcium, Vit B12.	avoid or prevent clinical deficiencies.
Caffeine	Coffee.	improve alertness, reduce fatigue.
Creatine	Creatine.	improve strength.
Dietary Nitrate/Sodium Bicarbonate	Beetroot juice	improve endurance efficiency.

Section 6: Caffeine

1. What are the main benefits of caffeine supplementation?

improve mental alertness & reaction time; Reduces perceived effort & fatigue.

2. In what types of sports/events is caffeine most effective?

Endurance events / Team and intermittent sports / High-intensity

3. What are possible side effects of caffeine supplementation?

sleep disturbance / Gastrointestinal upset.

4. What is the recommended dosage for caffeine supplementation (mg/kg)?

~~5-6 mg/kg body weight~~

Section 7: Creatine

1. What is creatine's main role in exercise performance?

To increase phosphocreatine store in muscle, improving ATP resynthesis

2. What form of creatine is best supported by evidence?

Creatine monohydrate.

3. In what sports or situations is creatine most beneficial?

Strength & power sports / Team sports / High intensity training blocks / Muscle building & rehab periods.

4. What are potential side effects or considerations when using creatine?

Possible stomach discomfort / weight gain.

Section 8: Dietary nitrate → 膳食硝酸盐

1. How does dietary nitrate improve exercise performance?

increase nitric oxide (NO) → improve blood flow → enhance endurance efficiency & time-to-exhaustion.

2. What are some common food sources of nitrate?

Beetroot / beetroot juice / Celery / Lettuce

3. In what types of events or sports might nitrate supplementation be most useful?

Endurance sports.

4. What are possible side effects or limitations of nitrate supplementation?

GI discomfort / Beetroot may cause red urine.

Section 9: Sodium bicarbonate → 碳酸氢钠

1. How does sodium bicarbonate act as a buffer during exercise?

Raises blood pH & increases buffering capacity.

2. What types of sports/events might benefit most from sodium bicarbonate use?

Swimming sprint (high-intensity events).

3. What are common side effects, and how can they be minimised?

bloating, diarrhoea // Take food, use chronic loading (0.3g/kg spread over the day).

Section 10: Reflection

- One new insight I gained today: _____
- How I will apply this knowledge in my practice: _____
- A supplement or dietary assessment tool I would like to research further: _____

Lecture 4: Disordered eating and eating disorders in athletes

Section 1: Disordered Eating vs Eating Disorders

1. What is the difference between disordered eating and a clinical eating disorder?

Disordered eating = unhealthy eating behaviours

A clinical diagnosed mental illness.

2. List three behaviours associated with disordered eating.

Food restriction / emotional eating / ~~over~~ excessive exercise.

3. What is the estimated prevalence of disordered eating in male and female athletes?

6-45% / 0-19% / 6-45% ^{endurance sports}
highest in weight-dependent

Section 2: Health & Performance Consequences

1. What are three potential health consequences of disordered eating?

hormonal dysfunction / gastrointestinal problems / poor immune function.

2. What are three performance-related consequences for athletes?

Reduced strength, endurance, power / Increased injury risk / Impaired concentration.

Section 3: Eating Disorders in Athletes

1. How does the DSM-5 define eating disorders?

Mental health disorders involving persistent disturbance in eating or eating-related behavior

2. Describe the key traits of anorexia nervosa.

Significantly low body weight / Intense fear of gaining weight

3. What are common signs/symptoms of bulimia nervosa?

Recurrent binge eating / Vomiting, excessive exercise.

4. What does EDNOS stand for, and what does it include?

Eating Disorder Not Otherwise Specified / Includes individuals with significant eating disorder symptoms that do not meet full criteria for anorexia or bulimia.

5. What is ARFID, and how does it differ from other eating disorders?

Avoidant / Restrictive food intake Disorder : Not based on body image concerns, unlike anorexia/bulimia; fear of vomit.

6. Define binge eating disorder, orthorexia nervosa, and bigorexia.

binge eating disorder: Recurrent binge eating without compensatory behaviours feeling of loss of control.
orthorexia nervosa: Obsession with "clean" or "healthy" eating.
bigorexia: Obsession with being more muscular.

Section 4: REDs (Relative Energy Deficiency in Sport)

1. What is REDs, and what are common causes in athletes?

Relative Energy Deficiency in Sport → pressure to be lean
Fear of weight gain

2. Why is diagnosing REDs challenging?

Athletes may appear "fit" / may hide restrictive behaviours.

3. What are some common signs/symptoms in female athletes? Male athletes?

Female : Hair loss, Mood change, Irregular periods.
Males : Low testosterone, Fatigue, low recovery

4. What is hypothalamic amenorrhoea (HA), and what are its consequences?

HA: Loss of menstrual periods due to low energy availability
Consequences: low bone density, Fatigue, poor performance, low estrogen.

Section 5: Case Studies

Case Study 1: A 17-year-old female distance runner reports irregular menstrual cycles, fatigue, and frequent stress fractures. Her diet log shows calorie restriction and avoidance of carbohydrates.

1. What are the red flags in this case?

1) Irregular menstrual cycles → possible HA 2) Frequent fractures → low bone density.
3) Restrictive eating / low-carb diet.

2. What nutritional and medical support might be needed?

Increase energy intake + carbohydrate focus / Education on fueling for performance.
Calcium + Vit D evaluation / Bone density test.

Case Study 2: A 22-year-old male rower is struggling with low libido, persistent fatigue, and recurrent illness. He reports training twice daily but not increasing calorie intake.

1. What might be happening here?

possible low testosterone / bulimia nervosa

2. How would you approach this situation as a sport rehabilitator?

Reset eating behaviour. / Encourage increased cal intake / monitor training.

Section 6: Support & Resources

1. What are the main organisations or charities that athletes can be referred to for eating disorder support?

BEAT (UK eating disorder charity) / NEDA (US)

2. Why is early identification and referral important?

Reduce risk for bone loss, hormonal damage

Section 7: Reflection

- One important warning sign I learned today: _____
- How I might support an athlete in this situation: _____
- Resources I would use to learn more: _____